

Milestone 1 Report

Advanced Threat Incident Analytics and Security Visualization Framework

1. Objective

The objective of Milestone 1 was to establish the foundation of the **Advanced Threat Incident Analytics and Security Visualization Framework** by integrating, cleaning, transforming, and modeling incident-related datasets in Power BI. The milestone focused on preparing reliable data for analysis, standardizing incident information, creating meaningful dimensions and relationships, and developing the initial analytical dashboard to support incident monitoring and security analysis.

2. Data Preparation and Modeling

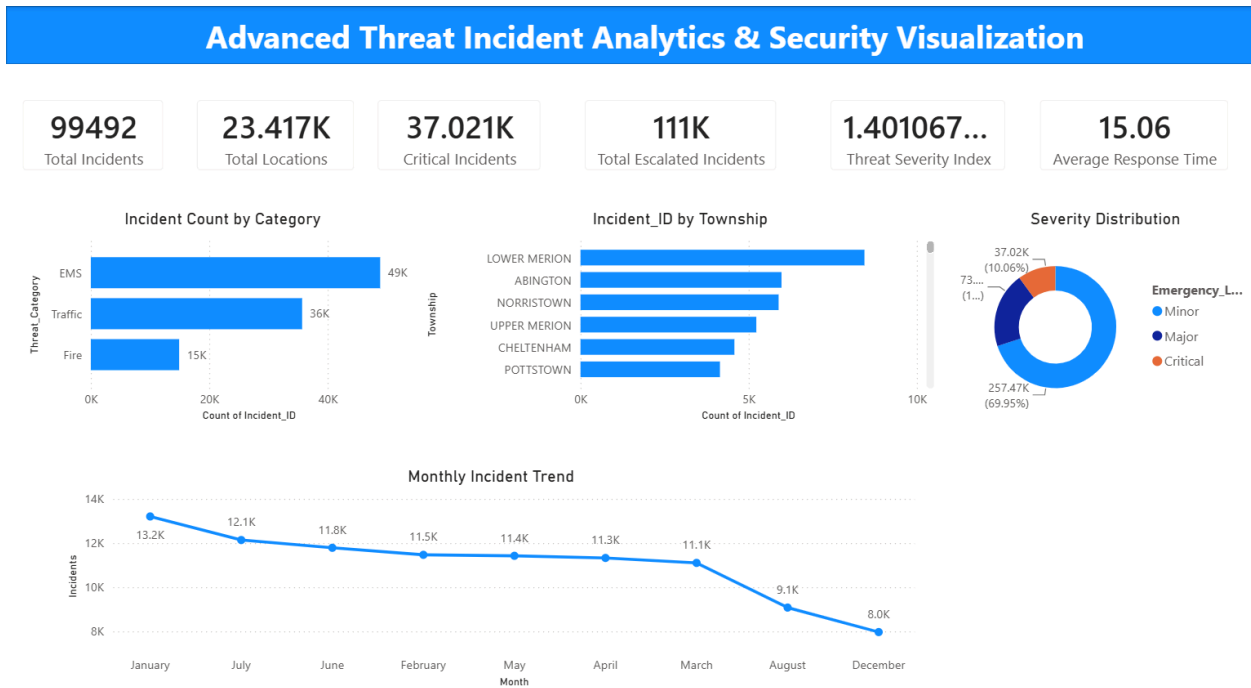
Multiple incident, location, weather, and response-related datasets were processed and standardized before visualization. The data-cleaning process included removing duplicate records, handling missing and inconsistent values, standardizing fields, and deriving useful attributes such as **Threat Category, Incident Type, Date, Month, Day, Hour, Season, and Weekend/Weekday classification**.

A structured data model was then created using a **fact-and-dimension approach**. The main **Fact_Incident** table was connected with dimension tables such as **Dim_Date, Dim_Location, Dim_Weather, and Dim_Response** using appropriate keys. This structure improved data organization and allowed different analytical perspectives to be combined within Power BI.

3. Dashboard

The Milestone 1 dashboard provides an initial overview of the incident dataset through KPI cards and visualizations for **incident volume, severity, categories, locations, and response-related information**. It establishes the foundation for the geographical and temporal analysis developed in the later milestones.

Dashboard Screenshot:



Link: https://drive.google.com/file/d/12mVo2Ztjo0wtjGw6YL3KjpLdLESFrz7Z/view?usp=drive_link

4. Key Insights

Insight 1 — Incident Distribution:

The dashboard provides an overall view of incident volume and demonstrates that incidents are distributed across multiple categories and types, making classification necessary for meaningful security analysis.

Insight 2 — Severity Analysis:

Incidents can be differentiated according to their severity levels, allowing critical and major incidents to be distinguished from lower-severity events and supporting prioritization of emergency response.

Insight 3 — Geographical Concentration:

Incident records show geographical concentration across different locations. Identifying these areas provides the foundation for detecting potential hotspots and prioritizing monitoring and response resources.

5. Business Recommendations

Recommendation 1 — Prioritize High-Severity Incidents:

Organizations should prioritize monitoring and response resources for Critical and Major incidents because these events represent greater potential operational impact.

Recommendation 2 — Strengthen Location-Based Monitoring:

Locations with comparatively high incident concentration should receive increased monitoring and preparedness measures. Geographical analysis can be used to support resource placement and improve response planning.